

Syed Mohsin Bukhari

AI Engineer | Machine Learning Researcher | Computer Vision Specialist

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Skills

AI & Deep Learning	📖	CNNs, RNNs, Transformers, image segmentation, object detection, multi-task learning.
ML Frameworks	📖	TensorFlow, PyTorch, Keras, scikit-learn, NumPy, Pandas, OpenCV, ONNX.
LLM & NLP	📖	Large Language Models, prompt engineering, LLM API integration, NLP, Multimodal AI.
Deployment	📖	Real-time inference, Edge AI, Jetson Nano, FPGA, Docker, REST APIs, Vercel, model integration.
Programming	📖	Python, JavaScript, Embedded C.
Tools	📖	Git, GitHub, Postman, Google Colab, Jupyter, KiCAD, Xilinx Vivado.

Education

2024 – 2026	📖	M.Tech – Artificial Intelligence and Machine Learning , Lovely Professional University, Punjab, India. CGPA: 7.3. Coursework: Machine Learning, Deep Learning, Computer Vision, Neural Networks, Data Analytics, Statistical Modeling.
2019 – 2023	📖	B.Tech – Electronics Engineering , Islamic University of Science and Technology, Awantipora, J&K. CGPA: 7.14. Foundation in signal processing, FPGA programming, embedded systems, and hardware–software integration.

Research and Publications

June 2026	📖	A Review on Artificial Intelligence Techniques for Breast Cancer Detection: Advances, Challenges, and Future Directions. Published in <i>Archives of Computational Methods in Engineering</i> , 2026. Q1 Scopus Indexed. DOI: 10.1007/s11831-026-10705-5. • Conducted a systematic review of AI techniques for breast cancer detection and analysis, covering peer-reviewed studies published from 2022 to 2024. • Analyzed 62 publications on classical machine learning, deep learning, hybrid AI, and explainable AI approaches for improving diagnostic precision and reducing misdiagnosis. • Reported CNN accuracy up to 99.43%, identified key clinical validation challenges, and discussed future directions including multimodal fusion, federated learning, and thermal imaging.
May 2025	📖	Real Time AI in Surgery: A Review of Precision, Innovation, and Future Directions in Surgical Assistance. International Journal of Performability Engineering (IJPE), May 2025. Q3 Scopus Indexed. DOI: 10.23940/ijpe.25.05.p2.249258. • Conducted a systematic review of state-of-the-art AI and ML techniques, including Computer Vision, CNN-based segmentation, and multimodal models for real-time intraoperative decision support. • Analyzed precision, latency, and safety trade-offs across AI-driven surgical assistance models, synthesizing future directions for surgical robotics and computer-assisted intervention. • Referenced leading architectures including Mask R-CNN, YOLACT, and Transformer-based models for surgical instrument detection and workflow recognition.

AI/ML Projects

ComplicationWatch

Real-Time Surgical Complication Precursor Detection Using Phase-Conditioned Normalizing Flows

Research Project | M.Tech Thesis Extension

Python, PyTorch, ResNet-50, RealNVP, Normalizing Flows, Computer Vision, Surgical AI, TensorRT, Jetson AGX Orin, Conformal Prediction.

- Designed an AI-powered surgical safety system to detect intraoperative complication precursors from laparoscopic video streams in real time.
- Built a phase-conditioned anomaly detection framework using ResNet-50 and RealNVP normalizing flows, modeling normal workflow across seven cholecystectomy phases without complication-labeled training data.
- Integrated temporal spike detection, conformal prediction, and TensorRT INT8 deployment, achieving AUROC 0.912, precision 0.843, recall 0.831, and 41.3-second average early warning time.

Visionary Fruit Sort

Production Computer Vision System

visionaryfruitsort.vercel.app

Python, CNN, TensorFlow, OpenCV, Jetson Nano, FPGA, Xilinx Vivado, KiCAD, React.js; Best Project of the Year.

- Designed and trained a CNN-based model for automated fruit quality grading and sorting under real-time inference conditions.
- Deployed on Jetson Nano edge hardware with FPGA acceleration, demonstrating production-grade Edge AI with custom PCB design.
- Built and deployed a full-stack web interface connecting hardware sensor input to a Vercel-hosted dashboard output.

Professional Experience

Nov 2023 – Apr 2024

Web Development Intern, EnableNow Technology Solutions, Srinagar, J&K.

- Integrated machine-learning-powered features into production web systems, bridging AI research with real-world software deployment.
- Developed and maintained applications serving 500+ daily active users and improved system performance by resolving backend service and API bottlenecks.
- Collaborated cross-functionally to deliver data-driven feature integrations, gaining production-level ML model deployment experience.

Achievements

2025

- **Peer-reviewed publication** in AI-driven surgical systems in the internationally indexed International Journal of Performability Engineering.

2023

- **Best Project of the Year** for a production-grade CNN-based computer vision system deployed on edge hardware.

Research

- Hands-on MICCAI-level surgical AI research experience with AutoLaparo, Mask R-CNN, YOLACT, and Transformer models.

Engineering

- End-to-end ML engineering across data preprocessing, model training, GPU optimization, edge deployment, and web-integrated inference pipelines.